

Scientific computing for geophysical problems

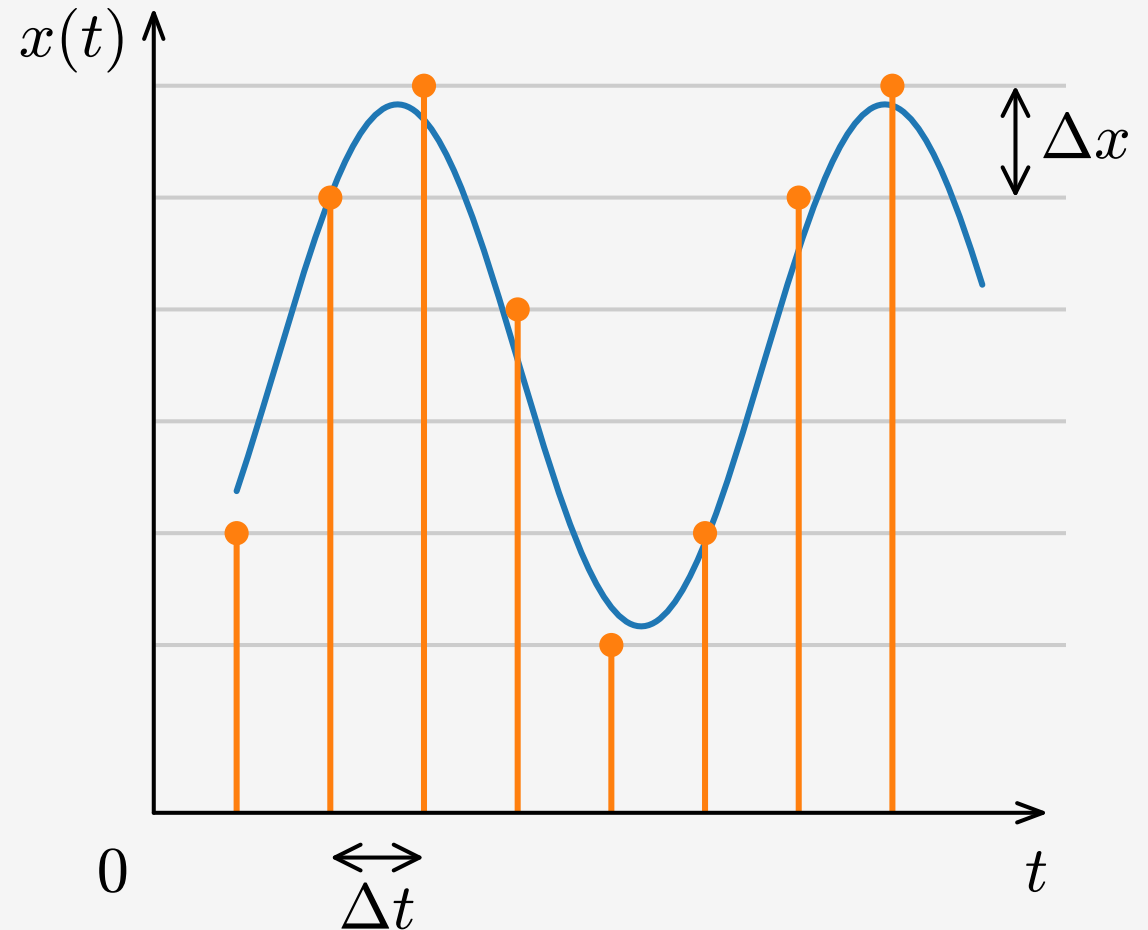
Digital signal processing

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Analog and digital signals

- The world is **analog**: ground motion, magnetic field, or sound are continuous
- A computer only stores a finite list of numbers $x[n]$.
Digitizing: sampling in time (at $n\Delta t$) and amplitude (at $m\Delta x$)

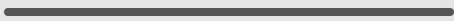


Digital signal processing in music

Piano record

$N = 24$ bits

$f_c = 44.1$ kHz

▶ 0:00 / 0:05  🔊 ⋮

$m = 16.7$ M levels

file size 1.6 MB

Piano record

$N = 8$ bits

$f_c = 8$ kHz

▶ 0:00 / 0:

$m = 256$ levels

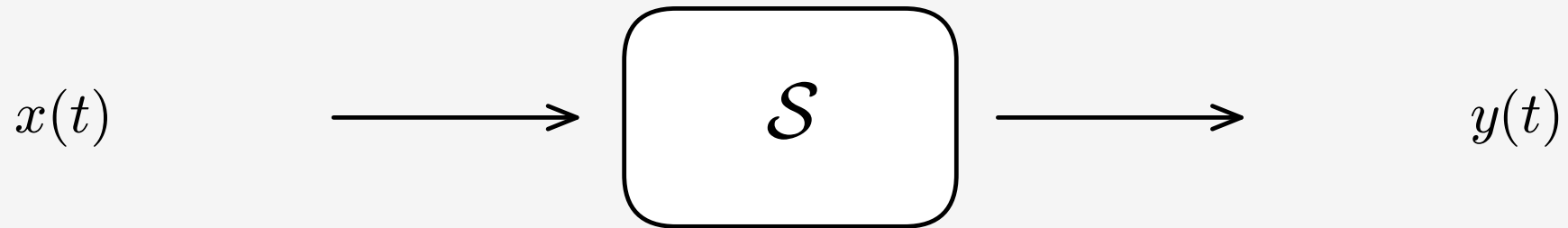
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16 times smaller, but audio quality is low.

Resolution and sampling rate both matter

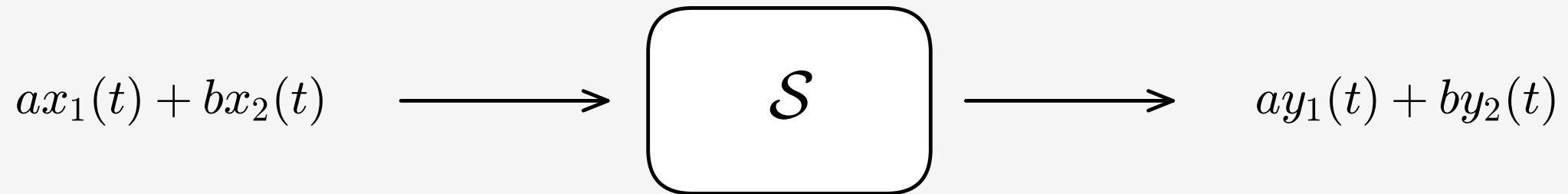
Signal processing as a system

A **system** transforms an input signal into an output signal:
a seismometer, a digital filter, or the Earth itself



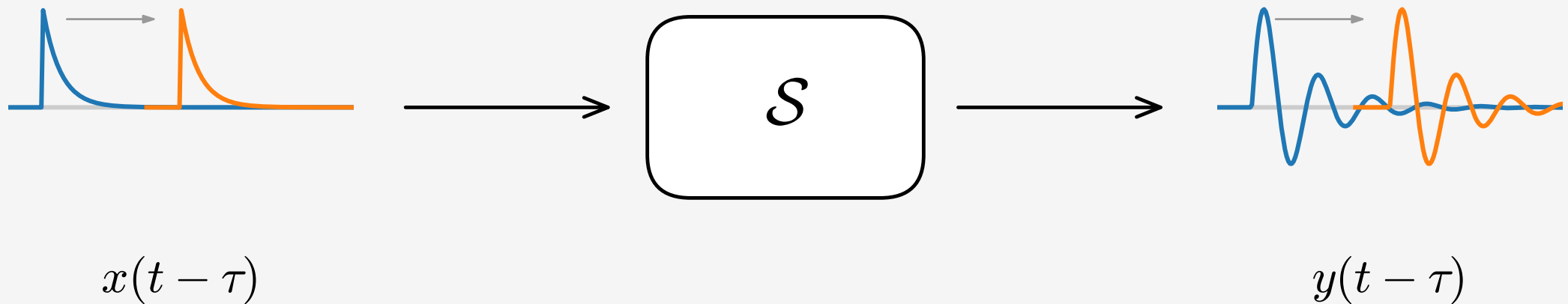
Linear systems

Linear: the response to a sum of inputs is the sum of their responses,
with $y_1 = \mathcal{S}\{x_1\}$ and $y_2 = \mathcal{S}\{x_2\}$



Time-invariant systems

Time-invariant: the system behaves the same at any time,
so a delayed input gives the same output, delayed



Linear and time-invariant systems

A system that satisfies both properties is called **LTI**

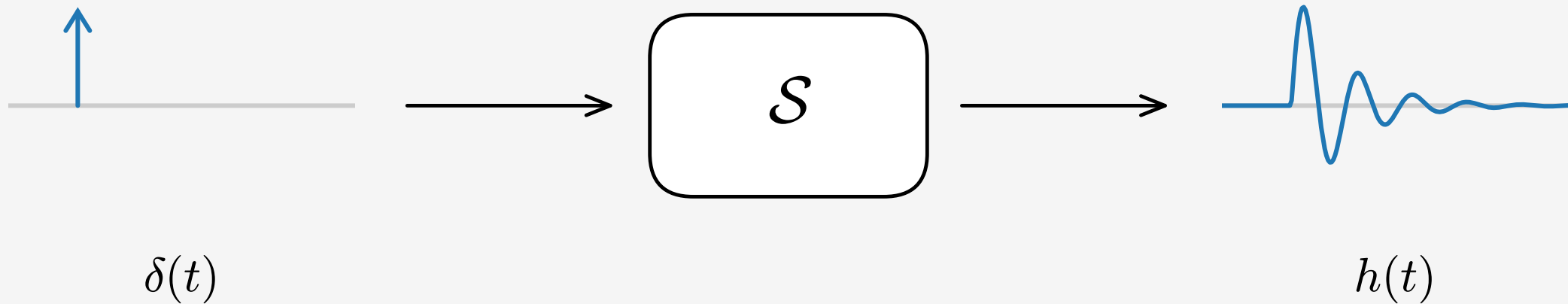
- Linearity: $\mathcal{S}\{ax_1(t) + bx_2(t)\} = ay_1(t) + by_2(t)$
- Time invariance: $\mathcal{S}\{x(t - \tau)\} = y(t - \tau)$

Seismic waves in the Earth, sensors, or digital filters are LTI to a good approximation

An **LTI** system is fully described by its response to a single impulse

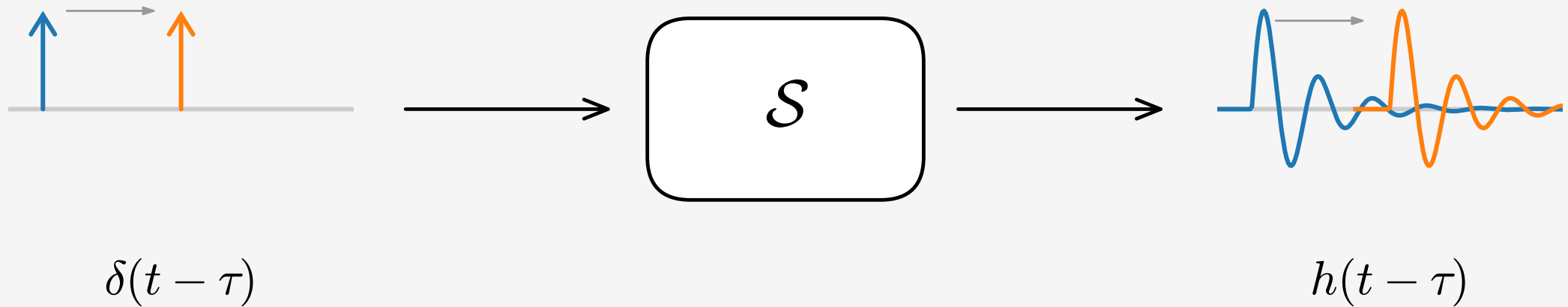
Impulse response

Hit the system with a very short pulse $\delta(t)$ and record the output:
this is the **impulse response** $h(t)$



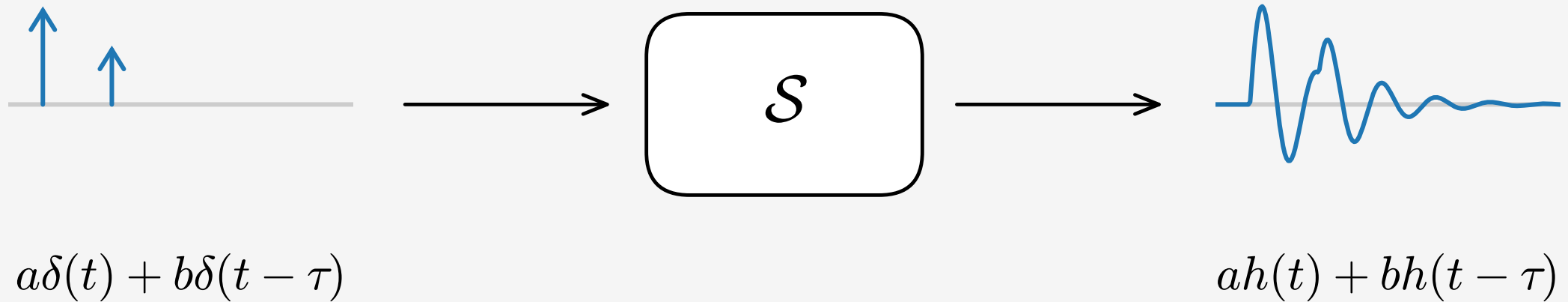
Impulse response and time invariance

Hitting later gives the same response, later



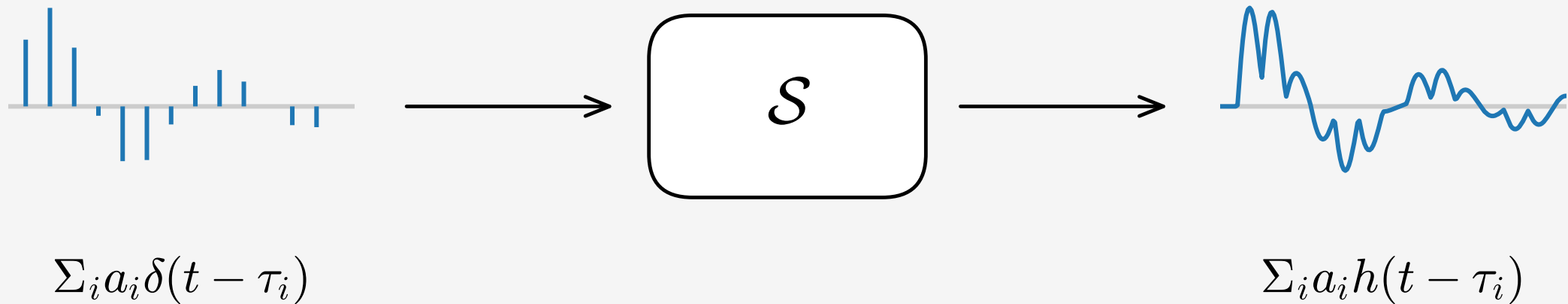
Impulse response and linearity

Two hits give the sum of two responses



Any signal is a sum of impulses

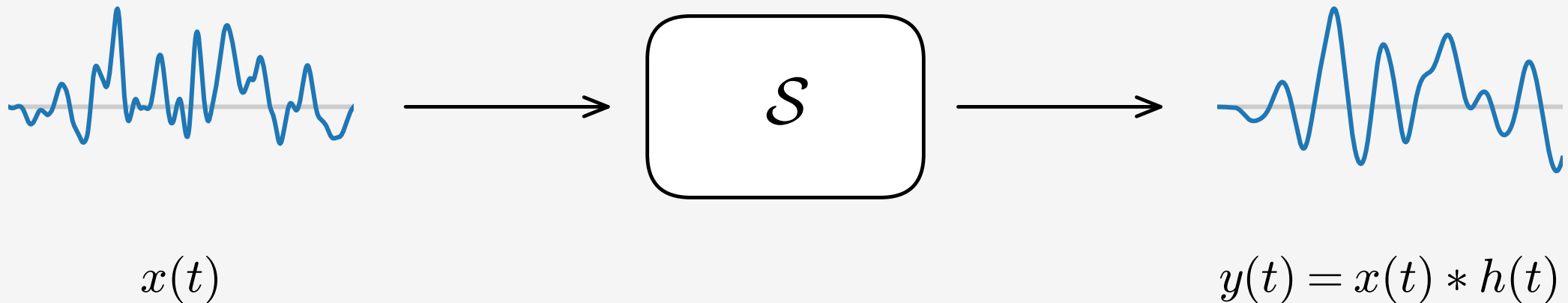
Any input can be seen as a series of small hits $a_i \delta(t - \tau_i)$ every $\Delta\tau$, with weights $a_i = x(\tau_i) \Delta\tau$, each producing its own response



Convolution

When the hits get infinitely close ($\Delta\tau \rightarrow 0$), the sums become integrals

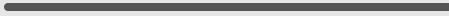
$$x(t) = \int_{-\infty}^{+\infty} x(\tau)\delta(t - \tau)d\tau \quad \Longrightarrow \quad y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau = (x * h)(t)$$



**The output of an LTI system is the convolution of the input
with h**

Example: adding reverb to a piano record

Hall impulse response $h(t)$
(a clap recorded in the hall)

▶ 0:00 / 0:07  🔊 ⋮

Dry piano record $x(t)$

▶ 0:00 / 0:

$$y(t) = (x * h)(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau$$

The piano, as if played in the hall

▶ 0:00 / 0:

Fourier transform

Any signal is a sum of sinusoids. The **Fourier transform** $\hat{x}(\omega)$ tells how much of each frequency $\omega = 2\pi f$ it contains, and can be inverted

$$\hat{x}(\omega) = \int_{-\infty}^{+\infty} x(t)e^{-i\omega t} dt \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} \hat{x}(\omega)e^{i\omega t} d\omega$$

- $|\hat{x}(\omega)|$ is the **spectrum**: the amplitude of each frequency
- The transform is linear: $\mathcal{F}\{ax_1 + bx_2\} = a\hat{x}_1 + b\hat{x}_2$

Convolution theorem

What does a convolution look like in the frequency domain? With the change of variable $u = t - \tau$, the double integral splits in two

$$\begin{aligned}\mathcal{F}\{x * h\}(\omega) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x(\tau) h(t - \tau) d\tau e^{-i\omega t} dt \\ &= \int_{-\infty}^{+\infty} x(\tau) e^{-i\omega\tau} d\tau \int_{-\infty}^{+\infty} h(u) e^{-i\omega u} du = \hat{x}(\omega) \hat{h}(\omega)\end{aligned}$$

Where $\hat{h}(\omega)$ is the frequency response. Filtering multiplies the spectrum by $\hat{h}$

Convolution in time is a product in frequency

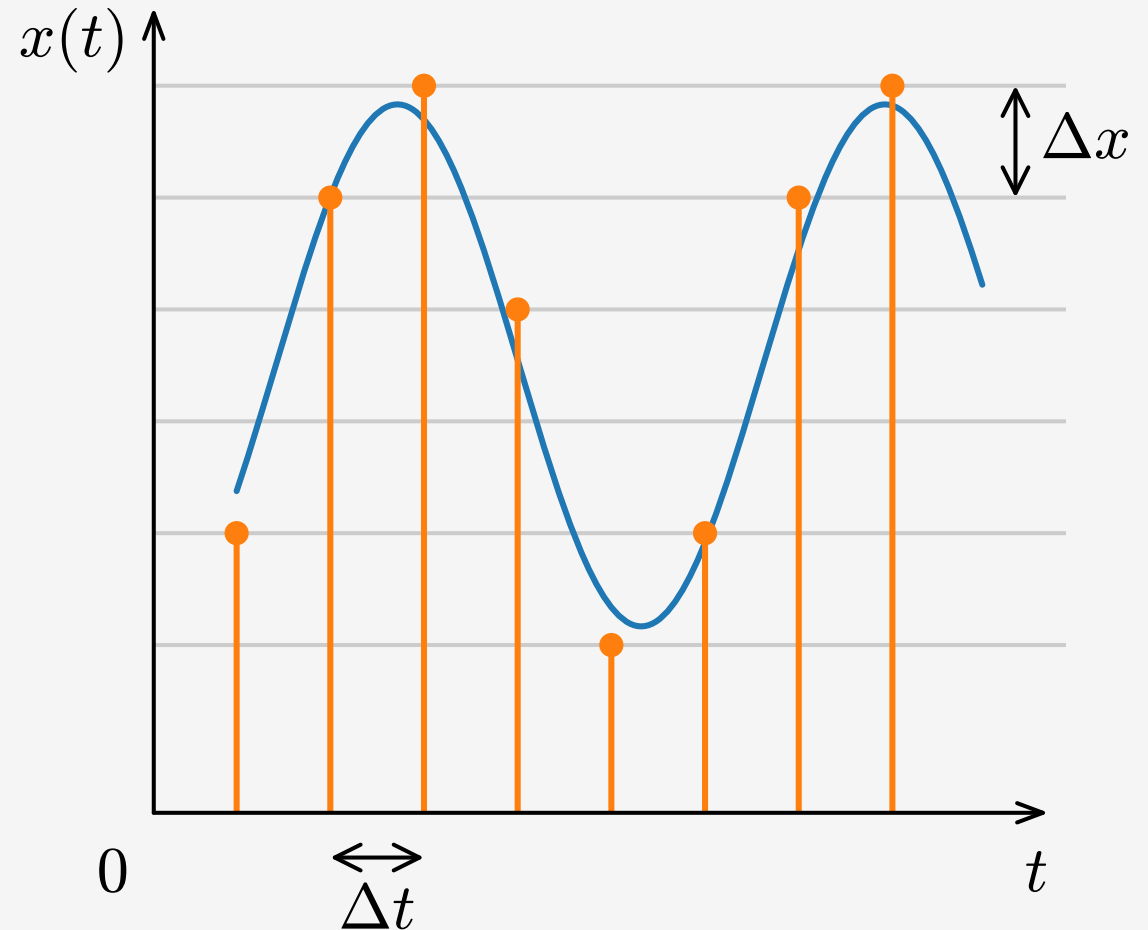
Sampling

Sampling keeps the values of $x(t)$ every Δt ; it multiplies the signal by a **Dirac comb** $\mathcal{W}_{\Delta t}$

$$\mathcal{W}_{\Delta t}(t) = \sum_{n=-\infty}^{+\infty} \delta(t - n\Delta t)$$

$$x_s(t) = x(t)\mathcal{W}_{\Delta t}(t)$$

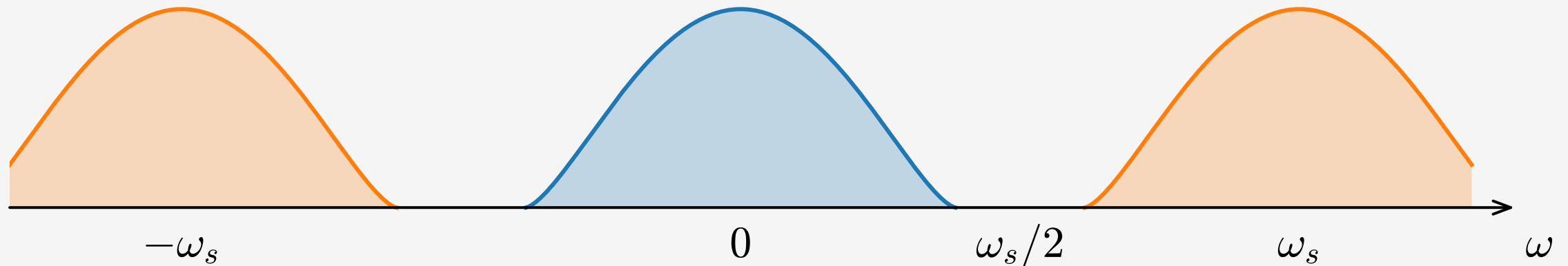
with $x[n] = x(n\Delta t)$ and the sampling rate $f_s = \Delta t^{-1}$.



Spectrum of a sampled signal

The Fourier transform of a comb is a comb, and by the convolution theorem, the product in time becomes a convolution in frequency

$$\hat{x}_s(\omega) = \frac{1}{2\pi} (\hat{x} * \hat{\mathcal{M}}_{\Delta t})(\omega) = \frac{1}{\Delta t} \sum_{k=-\infty}^{+\infty} \hat{x}(\omega - k\omega_s)$$



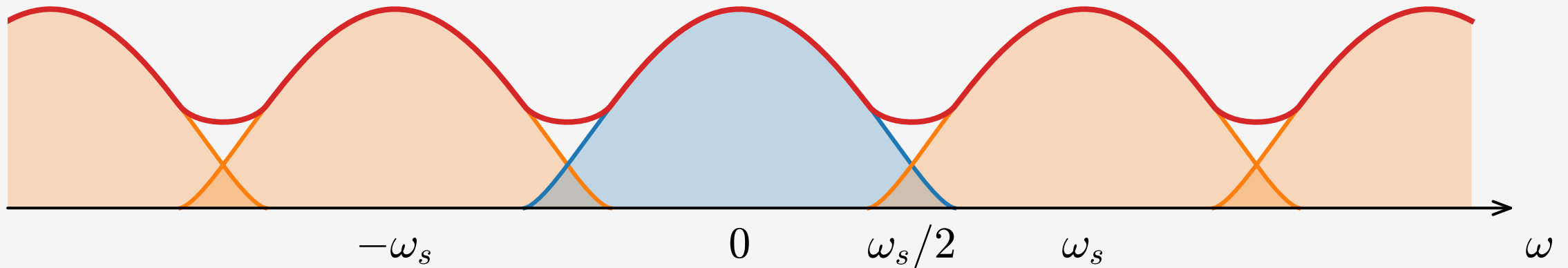
Sampling copies the spectrum at every multiple of ω_s

Aliasing

If $\omega_s < 2\omega_{max}$, the copies overlap:

high frequencies disguise as low ones, this is **aliasing**

Sampling theorem: a signal can be reconstructed if $\omega_s > 2\omega_{max}$,
that is, if its highest frequency is below the Nyquist frequency $f_s/2$



In practice, low-pass filter the signal before sampling it

Discrete-time Fourier transform

On a computer, we only have the samples $x[n]$.

Their Fourier transform is the **discrete-time Fourier transform** (DTFT)

$$X(\Omega) = \sum_{n=-\infty}^{+\infty} x[n]e^{-i\Omega n}$$

with the frequency $\Omega = \omega\Delta t$ in radians per sample, so that $X(\omega\Delta t) = \hat{x}_s(\omega)$

- $X(\Omega)$ is 2π -periodic, as the spectrum of the sampled signal
- The discrete convolution $(x * h)[n] = \sum_k x[k]h[n - k]$ becomes a product:
 $\mathcal{F}\{x * h\} = XH$

In practice: the discrete Fourier transform

A computer handles N samples, so we use the **discrete Fourier transform**

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-2i\pi kn/N}$$

- Computed efficiently with the fast Fourier transform (`numpy.fft`)
- Frequencies from 0 to $f_s/2$, every $1/(N\Delta t)$

Hands-on time

